

WIPRO Placement Paper 8

Written Test

This is a WIPRO model placement paper for placement preparation. This placement paper will cover aptitude, logical and reasoning, verbal and coding questions that are asked in WIPRO recruitment drives and also strictly follows the pattern of questions asked in WIPRO interviews. It is recommended to solve each one of the following questions to increase your chances of clearing the WIPRO interview.

Quantitative Analysis

1. What is the sum of two consecutive odd numbers, the difference of whose squares is 56?
 - a. 30
 - b. 28
 - c. 34
 - d. 32

Answer:

28

Explanation:

Let the no. be x and $(x + 2)$. Then $(x + 2)^2 - x^2 = 56$
 $4x + 4 = 56$
 $x + 1 = 14$
 $x = 13$
Sum of numbers = $x + (x + 2) = 28$

2. Express 252 as a product of primes.

- a. $2 * 2 * 3 * 3 * 7$
- b. $3 * 3 * 3 * 3 * 7$
- c. $2 * 2 * 2 * 3 * 7$
- d. $2 * 3 * 3 * 3 * 7$

Answer:

$$2 * 2 * 3 * 3 * 7$$

3. Two numbers are in the ratio 3 : 5. If their L.C.M. is 75. what is sum of the numbers?

- a. 25
- b. 45
- c. 40
- d. 50

Answer:

40

Explanation:

1st number = $3x$ 2nd number = $5x$ LCM of $3x$ and $5x$ is $15x \Rightarrow 15x = 75 \Rightarrow x = 5$ sum = $15+25=40$

4. A person employed a group of 20 men for a construction job. These 20 men working 8 hours a day can complete the job in 28 days. The work started on time but after 18 days, it was observed that two-thirds of the work was still pending. To avoid penalty and complete the work on time, the employer had to employ more men and also increase the working hours to 9 hours a day. Find the additional number of men employed if the efficiency of all men is the same.

- a. 40
- b. 44
- c. 64
- d. 80

Answer:

44

Explanation:

Let the total work be 3 units and additional men employed after 18 days be 'x'. $\Rightarrow$ Work done in first 18 days by 20 men working 8 hours a day = $(\frac{1}{3}) \times 3 = 1$ unit $\Rightarrow$ Work done in last 10 days by $(20 + x)$ men working 9 hours a day = $(\frac{2}{3}) \times 3 = 2$ unit Here, we need to apply the

formula $M_1 D_1 H_1 E_1 / W_1 = M_2 D_2 H_2 E_2 / W_2$, where $M_1 = 20$ men $D_1 = 18$ days $H_1 = 8$ hours/day $W_1 = 1$ unit $E_1 = E_2 = \text{Efficiency of each man}$ $M_2 = (20 + x)$ men $D_2 = 10$ days $H_2 = 9$ hours/day $W_2 = 2$ unit So, we have $20 \times 18 \times 8 / 1 = (20 + x) \times 10 \times 9 / 2 \Rightarrow x + 20 = 64 \Rightarrow x = 44$ Therefore, additional men employed = 44

5. Three pipes A, B and C were opened to fill a tank. Working alone, A, B and C require 10, 15 and 20 hours respectively. A was opened at 7 AM, B at 8 AM and C at 9 AM. At what time the tank would be completely filled, given that pipe C can only work for 3 hours at a stretch, and needs 1-hour standing time to work again.

- a. 12 : 00 PM
- b. 12 : 30 PM
- c. 1 : 00 PM
- d. 1 : 30 PM

Answer:

12 : 30 PM

Explanation:

Let the capacity of the tank be LCM (10, 15, 20) = 60 $\Rightarrow$ Efficiency of pipe A = $60 / 10 = 6$ units / hour $\Rightarrow$ Efficiency of pipe B = $60 / 15 = 4$ units / hour $\Rightarrow$ Efficiency of pipe C = $60 / 20 = 3$ units / hour $\Rightarrow$ Combined efficiency of all three pipes = 13 units / hour Till 9 AM, A works for 2 hours and B work for 1 hour. $\Rightarrow$ Tank filled in 2 hours by A = 12 units $\Rightarrow$ Tank filled in 1 hour by B = 4 units $\Rightarrow$ Tank filled till 9 AM = 16 units $\Rightarrow$ Tank still empty = $60 - 16 = 44$ units Now, all three pipes work for 3 hours with the efficiency of 13 units / hour. $\Rightarrow$ Tank filled in 3 more hours = 39 units $\Rightarrow$ Tank filled till 12 PM = $16 + 39$ units = 55 units $\Rightarrow$ Tank empty = $60 - 55 = 5$ units Now, C is closed for 1 hour and these remaining 5 units would be filled by A and B working together with the efficiency 10 units / hour. $\Rightarrow$ Time taken to fill these remaining 5 units = $5 / 10 = 0.5$ hours Therefore, time at which the tank will be completely filled = 12 PM + 0.5 hours = 12 : 30 PM

6. Rajdhani Express halts for 3 minutes every time it covers a distance of 75 km. If the train runs at a speed of 100 km/h and the destination is 600 km away from the source, find out the time taken to reach the destination station from the source station.

- a. 6 h 21 min
- b. 6 h 22 min
- c. 6 h 23 min

d. 6 h 24 min

Answer:

6 h 21 min

Explanation:

*Since the train runs at a speed of 100 km/h, the time taken to cover 600 km = $600/100 = 6$ h. Number of times the train halts = $600/75 - 1 = 7$. Since the train halts for 3 minutes at each stop, the time spent waiting = $7 * 3 = 21$ min. Therefore, total time taken = 6 h 21 min.*

7. A train crosses a pole in 20 sec. If the length of train is 500 meters, what is the speed of the train?

- a. 27 m/s
- b. 20 m/s
- c. 25 m/s
- d. 30 m/s

Answer:

25 m/s

Explanation:

$$V = 500/20 = 25 \text{ m/s}$$

8. Felix spends 66.66% of his salary and saves Rs 1200 every month. Calculate his monthly expenditure in Rupees.

- a. Rs 2402
- b. Rs 2400
- c. Rs 2401
- d. Rs 2405

Answer:

Rs 2400

Explanation:

Let Felix's monthly salary be Rs x . Then, $(100 - 66.66)\%$ of $x = \text{Rs } 1200 \Rightarrow 33.33\%$ of $x = \text{Rs } 1200 \Rightarrow x/3 = \text{Rs } 1200 \Rightarrow x = \text{Rs } 3600$. Therefore, his monthly expenditure = $3600 - 1200 = \text{Rs } 2400$.

9. Mark, Steve and Bill get their salaries in the ratio of 2:3:5. If their salaries are incremented by 15%, 10%, and 20% respectively, the new ratio of their salaries becomes:

- a. 8:16:15
- b. 23:33:60
- c. 33:30:20
- d. 21:25:32

Answer:

23:33:60

Explanation:

Let their old salaries be $2a$, $3a$, $5a$ respectively. Then, their new salaries become: 115% of $2a = 2a \times 1.15 = 2.3a$ 110% of $3a = 3a \times 1.10 = 3.3a$ 120% of $5a = 5a \times 1.20 = 6a$ So, the new ratio becomes $2.3a:3.3a:6a$ Upon simplification, this becomes $23:33:60$

10. The present ages of A, B and C are in proportions 4:5:9. Nine years ago, sum of their ages was 45 years. Find their present ages in years

- a. 15, 20, 35
- b. 20, 24, 36
- c. 20, 25, 45
- d. 16, 20, 36

Answer:

16, 20, 36

Explanation:

Let the current ages of A, B and C be ax years, $5x$ years and $9x$ respectively. Then $(4x-9) + (5x-9) + (9x-9) = 45 \Rightarrow 18x - 27 = 45 \Rightarrow 18x = 72 \Rightarrow x = 4$ Present ages of A, B and C are $4x = 16$, $5x = 20$, $9x = 36$ respectively.

Verbal Reasoning

1. We _____ our friend's birthday and we _____ how to make it up to him.
- a. Completely forgot --- don't just know
 - b. Forgot completely --- don't just know
 - c. Completely forgot --- just don't know
 - d. Forgot completely --- just don't know

Answer:

Completely forgot --- just don't know

Explanation:

We Completely forgot our friend's birthday and we just don't know how to make it up to him.

2. Choose the statement where underlined word is used correctly.
- a. The industrialist had a personnel jet
 - b. I write my experience in my personnel diary
 - c. All personnel are being given the day off
 - d. Being religious is a personnel aspect

Answer:

All personnel are being given the day off

Explanation:

personnel mean a body of persons employed in an organization or place of work.

3. A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is
- a. fabric
 - b. textile
 - c. fibre
 - d. apparel

Answer:

fibre

Explanation:

apparel means clothing.

4. **Out of the following four sentences, select the most suitable sentence with respect to grammar and usage:**
- a. Since the report lacked needed information, it was of no use to them
 - b. The report was useless to them because there were no needed information in it
 - c. Since the report did not contain the needed information, it was not real useful to them
 - d. Since the report lacked needed information, it would not had been useful to them

Answer:

Since the report lacked needed information, it was of no use to them

Explanation:

*Only option (A) seems correct. In option (B) - **were** is wrong used here. In option (C) - **real** is wrong, should be really In option (D) - **had** is wrong. would not have been is correct.*

5. **The Tamil version of _____ John Abraham-starrer Madras Cafe _____ cleared by the Censor Board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu _____ this Friday.**
- a. Mr., was, found, on
 - b. a, was, found, at
 - c. the, was, found, on
 - d. a, being, find at

Answer:

the, was, found, on

6. **Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.**

- a. related
- b. extraneous
- c. outside
- d. useful

Answer:

extraneous

7. Select the pair that best expresses a relationship similar to that expressed in the pair: Children : Pediatrician

- a. Adult : Orthopaedist
- b. Females : Gynaecologist
- c. Kidney : Nephrologist
- d. Skin : Dermatologist

Answer:

Females : Gynaecologist

Explanation:

Pediatrician is a doctor for children. Gynaecologist is a doctor for Females.

8. Select the correct SYNONYM for: savage

- a. barbaric
- b. new
- c. same
- d. nice

Answer:

A

9. Select the correct SYNONYM for: Feral

- a. Crafty

- b. Nearly
- c. Relevant
- d. Ferocious

Answer:

D

10. Select the correct ANTONYM for: Physical

- a. Substantial
- b. Immaterial
- c. Real
- d. Natural

Answer:

B

Logical Reasoning

1. Find wrong number in series: 8, 24, 12, 36, 18, 54, 26

- a. 12
- b. 24
- c. 18
- d. 26

Answer:

15

Explanation:

*Two alternate series: $8 * 3 = 24$ $24 / 2 = 12$ $12 * 3 = 36$ $36 / 2 = 18$ $18 * 3 = 54$ $54 / 2 = 27$*

2. Find wrong number in series: 7, 26, 63, 124, 215, 342, 496

- a. 7
- b. 63

c. 215

d. 496

Answer:

496

Explanation:

$$(2^3 - 1), (3^3 - 1), (4^3 - 1), (5^3 - 1), (6^3 - 1), (7^3 - 1), (8^3 - 1) = 511$$

Directions to solve Question 3 and 4: Each of the questions given below consists of a statement and/or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the given question.

Read both the statements and Give answer

(a) if the data in Statement I alone is sufficient to answer the question, while the data in Statement II alone is not sufficient to answer the question.

(b) if the data in Statement II alone is sufficient to answer the question, while the data in Statement I alone is not sufficient to answer the question.

(c) if the data in each Statement I and Statement II alone is sufficient to answer the question.

(d) if the data even in both Statements I and II together are not sufficient to answer the question.

(e) if the data in both Statements I and II together are necessary to answer the question.

3. What is the probability that $x^3 - 8 = 0$ when x is selected from a set of 8 integers?

I. The smallest number in the set is -11

II. The arithmetic mean of the set is $1/8$.

a. (a)

b. (b)

c. (c)

d. (d)

e. (e)

Answer:

(d) if the data even in both Statements I and II together are not sufficient to answer the question.

4. How many children are there in a row? Statement I. A is tenth from the left in the row. Statement II. B is exactly in the middle and there are fifteen children towards his right.
- a. (a)
 - b. (b)
 - c. (c)
 - d. (d)
 - e. (e)

Answer:

(b) if the data in Statement II alone is sufficient to answer the question, while the data in Statement I alone is not sufficient to answer the question.

5. Consider the following phrase: **Statement:** Some S are L Some C are P Some P are R **Conclusions: I.** Some P are L **II.** Some C are R **Choose the correct option given below:**
- a. only conclusion I is true.
 - b. only conclusion II is true.
 - c. either conclusion I or conclusion II is true
 - d. neither conclusion I nor conclusion II is true
 - e. both conclusions I and II are true.

Answer:

(d)

6. Consider the following phrase: **Statement:** Fruits are getting costlier day by day **Assumptions: I.** Fruits are a rare commodity **II.** One should not eat Fruits Choose the correct option given below.
- a. If only assumption I is implicit
 - b. If only assumption II is implicit.

- c. If either I or II is implicit.
- d. If neither I nor II is implicit.
- e. If both I and II are implicit.

Answer:

(d)

7. Consider the following phrase: **Statement:** Cactus plants grow in deserts and requires little water **Assumptions:** I. All plants that grow in desert require little water II. Cactus plants might not grow in place with more water Choose the correct option given below.

- a. If only assumption I is implicit
- b. If only assumption II is implicit.
- c. If either I or II is implicit.
- d. If neither I nor II is implicit.
- e. If both I and II are implicit.

Answer:

(d)

8. **Statements:** I - Some S are L II - Some C are P III - All P is R **Conclusions:** I. Some P are L II. Some C are R **Choose the correct option given below:**

- a. only conclusion I is true.
- b. only conclusion II is true.
- c. either conclusion I or conclusion II is true
- d. neither conclusion I nor conclusion II is true
- e. both conclusions I and II are true.

Answer:

(d) neither conclusion I nor conclusion II is true

9. Find wrong number in series: 3, 15, 35, 63, 143

- a. 15
- b. 35
- c. 63
- d. 143

Answer:

143

Explanation:

*Series: $S_n = 1*3 + 3*5 + 5*7 + \dots$*

10. Find wrong number in series: -1, 8, -27, 64, 125

- a. 8
- b. -27
- c. 64
- d. 125

Answer:

125

Explanation:

$T_n = -1^3 + 2^3 - 3^3 + 4^3 - \dots$